

Empirical Assignment

ECON 300: Labour Economics

Due: April 07, 2026

Objective

This assignment applies human capital theory and compensating wage differentials using real data.

Dataset

You are provided with a dataset containing:

- Wage
- Education
- Experience
- Job risk indicator

Part I: Descriptive Analysis (10 marks)

1. Compute summary statistics.
2. Plot wage vs education.
3. Plot wage distribution.
4. Briefly describe patterns.

Part II: Human Capital Model (20 marks)

Estimate:

$$\ln(wage_i) = \beta_0 + \beta_1 education_i + \beta_2 experience_i + \beta_3 experience_i^2 + u_i$$

1. Interpret coefficients.
2. Explain diminishing returns.
3. Compute marginal effect of experience.

Part III: Compensating Differentials (20 marks)

Estimate:

$$\ln(wage_i) = \beta_0 + \beta_1 education_i + \beta_2 experience_i + \beta_3 risk_i + u_i$$

1. Interpret the coefficient on risk.
2. Does the data support compensating wage differentials?

Part IV: Policy Discussion (15 marks)

1. Should education be subsidized?
2. Should risky jobs be regulated?

Submission

Submit:

- Report (max 5 pages)
- Graphs and tables
- Code/Excel Sheet
- **Late Submission penalty: 10% per day.**